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1 October 1979
Superseding
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29 October 1958

MILITARY SPECIFICATION
GENERATOR, DIRECT CURRENT, G-43( )/G (HAND CRANKED)

This specification is approved for use by the Communications Electronics Materiel Readiness Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense

1. SCOPE

1.1 Scope.- This specification covers one type of hand-cranked, direct current generator designated as Generator, Direct Current, G-43( )/G.

2. APPLICABLE DOCUMENTS

2.1 The following specifications, standards, and drawings of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

FEDERAL

L-P-378	Plastic Sheet and Strip, thin Gauge, Polyolefin
NN-P-71	Pallet, Material Handling, Wood, Stringer Construction, 2 Way and 4 Way (Partial)
QQ-S-781	Strapping, Steel, Flat and Seals
PPP-B-585	Box, Wood, Wirebound
PPP-B-601	Box, Wood, Cleated Plywood
PPP-B-621	Box, Wood, Nailed and Lock Corner
PPP-B-636	Box, Shipping, Fiberboard
PPP-B-640	Box, Fiberboard, Corrugated, Triple-Wall
PPP-C-843	Cushing, Material, Cellulosic

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Communications Electronics Materiel Readiness Command, Fort Monmouth, New Jersey 07703 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6115

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PPP-F-320	Fiberboard, Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes
PPP-S-760	Strapping, Nonmetallic, (And Connectors)
PPP-T-45	Tape, Gummed, Paper, Reinforced and Plain, for Sealing and Securing
PPP-T-97	Tape, Pressure-sensitive Adhesive, Filament Reinforced

MILITARY

MIL-P-116	Preservation-packaging, Methods of
MIL-B-117	Bag, Sleeve and Tubing - Interior Packaging
MIL-M-13231	Marking of Electronic Items
MIL-F-14072	Finishes for Ground Electronic Equipment

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-147	Palletized and Containerized Units Loads 40 inch x 48 inch Pallets, Skids, Runners or Pallet Type Base
MIL-STD-169	Extreme-temperature Cycle
MIL-STD-170	Moisture Resistance Test Cycle for Ground Signal Equipment
MIL-STD-252	Wired Equipment, Classification of Visual and Mechanical Defects
MIL-STD-461	Electromagnetic Interference Characteristics Requirements for Equipment
MIL-STD-462	Electromagnetic Interference Characteristics, Measurements of
MIL-STD-1188	Commercial Packaging of Supplies and Equipment

DRAWINGS

SIGNAL CORPS

SC-DL-202175	Generator, Electrical G-43( )/G Drawing and Data List

(Copies of specifications, standards and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer. Both the title and number or symbol should be stipulated when requesting copies).

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3. REQUIREMENTS

3.1 Construction.- Generator, Direct Current G-43( )/G shall be constructed in accordance with Drawing and Data List SC-DL-202175.

3.2 Plastic material and parts.- Where not machined, plastic material and parts shall have the original smooth or polished surfaces. Surfaces that have been sawed, cut, punched, or otherwise machined shall be as smooth as practicable in accordance with good manufacturing practice for the intended application.

3.3 Wiring and Cabling.- Wiring and cabling shall be neat and sturdy.

3.3.1 Slack.- Wires and cables shall be as short as practicable except that sufficient slack shall be provided to prevent undue stress on cable forms, wires, and connections; to enable parts to be removed and replaced during servicing without disconnecting other parts; and to facilitate field repair of broken or cut wires.

3.3.2 Protection.- Wires and cables shall be so placed and protected as to avoid contact, under specified service conditions, with rough or irregular surfaces or sharp edges. Wires shall not be bent sharply where they enter insulation material. Where wires run through holes or slots in metal (partitions, shields, and similar items), they shall be protected by suitable grommets or bushings.

3.3.3 Support.- Wires and cables shall be properly supported and secured, to prevent undue stress on conductors and terminals and undue change in position of the wires and cables after the equipment has been subjected to specified service conditions or has been serviced or repaired in a normal manner.

3.3.4 Clearance.- Clearance between solder connections or bare conductors, on terminal boards, relays, or other parts, shall be at least 1/4 inch unless use of specified parts makes such clearance impracticable.

3.4 Soldering.-

3.4.1 Flux and cleaning agents.- No acid or acid salts shall be used in preparation for or during soldering; however, exception is permitted for preliminary tinning of electrical connections and for tinning or soldering of mechanical joints not used to complete electrical circuits, but in no case shall acid or acid salts be used where they can come in contact with insulation material. Where acid or acid salts are used, as permitted above, they shall be completely neutralized and removed immediately after use.

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3.4.2 Process.- There shall be no sharp points or rough surfaces resulting from insufficient heating. The solder shall feather out to a thin edge, indicating proper flowing and wetting action, and shall not be crystallized, overheated, or underheated. The minimum necessary amount of flux and solder shall be used for electrical connections. Any means employed to remove an unavoidable excess of flux shall not incur the risk of loose particles of flux, brush bristles, or other foreign material remaining in the equipment: flux being spread over a larger area; or damage to the equipment. Insulation material that has been subjected to heating during the soldering operation shall be undamaged and parts fastened shall not have become loosened.

3.5 Cleaning.- Metal parts, after fabrication, shall be cleaned in accordance with MIL-F-14072 or in accordance with good commercial practice. Cleaning processes shall have no deleterious effect on the equipment. Corrosive material shall be removed completely before parts are assembled into the equipment. After assembly, components shall be cleaned thoroughly and shall be free from particles of solder, flux, and other foreign material. If necessary, such cleaning shall also be performed before final assembly of components.

3.6 Riveting.- Thickness of countersunk metal which accepts the heads of flush rivets shall not be less than the height of the rivet heads. The distance from the centers of rivet holes to the edges of the material in which the holes are formed shall be at least 1 1/2 times the rivet diameter. Rivets shall be used where practicable in preference to other means of securing aluminum of magnesium structural parts if the joint is subjected to dynamic tension loads when the equipment is exposed to specified service conditions and if the parts do not require removal when the equipment is serviced. After riveting, the joints shall be tight, the joined parts shall be undamaged, and the rivet heads shall be properly seated and tight against their bearing surfaces. Riveting shall not be used for mounting parts such as capacitors and resistors.

3.7 Marking.- Marking shall conform to MIL-M-13231. External identification symbols shall be Group I as described in that specification.

3.7.1 Visibility.- Wherever practical, parts shall be so mounted that their identification markings will be readily visible with minimum disassembly of the equipment.

3.7.2 Serial Numbers.- The generators shall have serial numbers.

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3.8 Electrical requirements.- All electrical measurements shall be made at the output plug of the generator.

3.8.1 Output.- With continuous operation and a cranking speed between 50 to 70 rpm, the output of the generator shall be as follows:

VOLTAGE	CURRENT
425 ± 25	115 milliamperes
105	32 ± 3.25 milliamperes
6.3 plus 0.2 minus 0.1	2.5 amperes
1.4 plus 0.2	0.465 plus 0.035 amperes minus 0.0

3.8.2 Voltage regulation.-

3.8.2.1 High voltage.- The high voltage shall not vary more than 8 percent from its full load value when tested in accordance with 4.22.1.

3.8.2.2 Low voltage.- The low voltage output shall not vary more than 10 percent from its full load value when tested in accordance with 4.22.2.

3.8.2.3 Change of speed.- The output voltages shall not vary more than ± 3 percent when tested in accordance with 4.22.3.

3.8.3 Low voltage change with change of temperature.- With the generator operating at any cranking speed between 50 and 70 rpm, inclusive, and at any temperature between -40°F and +125°F, the low voltage shall not deviate more than 2 percent from the value measured under the same load and cranking speed at 68°F to 70°F temperature when tested in accordance with 4.19.1.

3.8.4 Ripple voltage .- The rms ripple shall not exceed 1 1/2 per cent of the low output voltage, or 1 1/4 per cent of the high output voltage when the generator is operating at any temperature between -40°F and +125°F.

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3.8.5 Efficiency.- The efficiency shall not be less than the values specified for the following ambient temperatures when tested in accordance with 4.10.

AMBIENT TEMPERATURE	EFFICIENCY
-40°F	45 percent
68°F - 70°F	60 percent
+ 125°F	55 percent

3.8.6 Temperature rise.- The temperature rise of the generating unit, at any ambient temperature between 50°F and 104°F, shall not exceed 54°F. (See 4.20).

3.8.7 Breakdown voltage.- Air gaps and electrical insulation shall have the minimum breakdown voltage shown in TABLE I. After 10-second application of the voltage, there shall be no visible deterioration and the equipment shall meet specified performance requirements. (See 4.9).

TABLE I. - Breakdown voltage

Operating voltage (Note A)	Breakdown voltage (Note B)
0 to 30 volts	400 volts
30 to 125 volts	500 volts
125 to 500 volts	4 times operating voltage
500 volts or more	1000 volts plus 2 times operating voltage

Note A: Operating voltage shall be maximum instantaneous ac plus dc during use of the equipment under specified service conditions.

Note B: Minimum breakdown voltage shall be dc or peak ac. If ac, the wave form shall be approximately sinusoidal and the frequency not over 100 cycles per second.

3.8.8 Commutation.- Commutation shall be successfully effected at all speeds and from no load to full load. Minor sparking will not be considered evidence of unsuccessfull commutation.

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3.9 Acoustic noise.- The acoustic noise produced by the generator shall be less than a sound level of 60 db at 70 db weighting, and 59 db at 40 db weighting when tested in accordance with 4.7.

3.10 Radio operation.- With the generator furnishing power to receiver-transmitter RT-77( )/GRC-9, the following requirements shall be met when tested in accordance with 4.15.

(a) The noise output of the receiver due to hum, ripple, and hash of the generator shall be such as to provide a signal plus-noise to noise ratio of 10 to 1 or greater with a 6 microvolt signal input at 10 milliwatts output.

(b) The no-signal radio frequency hash in the output of the receiver shall not exceed 0.3 milliwatt with no antenna connected and 0.5 milliwatt with an antenna connected.

(c) The transmitted CW, MCW or voice modulated signal shall show no evidence of ripple, warble, hash, or distortion.

3.10.1 Electromagnetic Interference (EMI).- The equipment shall comply with the requirements of MIL-STD-461A, Notice 4, Table A-II, Class IIIa, for reprocurements of equipment developed under MIL-I-11748. Only Method RE02 is applicable (see 4.16).

3.11 Life.- The generator shall be capable of meeting the electrical and mechanical requirements of the specification after a life test of 1000 hours when tested in accordance with 4.12.

3.12 Finish.- The equipment shall be finished in accordance with Specification MIL-F-14072 and the equipment drawings. (See 4.4).

3.13 Air-seal test, provision for.- The equipment shall provide means for enabling performance of the air-seal test. (See 4.5).

3.14 Service conditions.- The equipment shall meet the following service conditions:

3.14.1 Equipment nonoperating.- The equipment shall comply with the operational requirements of 3.14.2 after subjection to any of the following nonoperating conditions successively or in combinations encountered during worldwide short-term storage and transit:

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(a) Temperature.- Exposure in the range of +160°F to -80°F; however, exposure at the high temperature extreme not to exceed 4 hours and at the low temperature extreme not to exceed 24 hours at any one time. (See 4.19).

(b) Relative humidity.- Relative humidity up to 97 percent, for an indefinite period of time; and exposure at 100 percent relative humidity, with condensation, for 4 hours. (See 4.13).

(c) Immersion.- Three feet of water for 2 hours. (See 4.11).

3.14.2 Equipment operating.- The equipment shall meet the performance requirements of this specification while subjected to any of the following conditions successively or in combinations likely to be encountered during worldwide operations:

(a) Operation.- Continuous operation for a period of 1000 hours, alternating 2 Min. on and 4 Min. off, with no more than normal maintenance and replacement of parts. (See 4.12)

(b) Temperature.- Ambient temperature in the range of +125°F to -40°F. Exposure at the high temperature extreme not to exceed 4 hours and at the low temperature extreme not to exceed 72 hours. (See 4.19)

(c) Relative humidity.- Relative humidity up to 97 percent; and exposure at 100 percent relative humidity, with condensation, for 4 hours. (See 4.13)

3.15 Bounce and shock.- The equipment shall meet the requirements of Table II.

TABLE II .- Bounce and shock

Requirement	Inspection paragraph	Performance after inspection
Bounce	4.8	Full specification performance. No physical damage except surface abrasion.
Shock, drop	4.18	Equipment shall be operable. Minor mechanical damage is permissible (See note A)

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NOTE A: After this shock test, equipment performance shall conform to the following:

3.8.1 Output rating
3.8.3 Voltage change with change of temperature
3.8.5 Overall efficiency
3.9 Acoustical noise
3.10 Radio operation

3.16 First Article Samples.- The contractor shall furnish First Article samples for approval, as required by the invitation for bids and contract. (See 6.2(c))

3.17 Technical literature, tools, and running spare parts.- Technical literature, tools, and running spare parts shall be furnished as specified in the contract. Running spare parts shall be identical to corresponding parts in the items furnished on the order. (See 6.2)

3.18 Workmanship.- The item shall be manufactured and assembled in accordance with the applicable portions of the following paragraphs herein:

3.2 General requirements for plastic material and plastic parts.
3.3 Wiring and cabling.
3.3.1 Slack in wires and cables.
3.3.2 Protection for wires and cables.
3.3.4 Clearance in wiring and cabling.
3.4.1 Flux and cleaning agents for soldering.
3.4.2 Process for soldering.
3.5 Cleaning of equipment.
3.6 Riveting.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

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4.1.1 Classification of inspection.- Inspection shall be classified as follows:

(a) First Article Inspection (does not include preparation for delivery). (See 4.2)

(b) Procurement inspection. (Procurement inspection shall be the inspection performed by the contractor and by the Government, as specified by 4.3 and 4.23)

(1) Procurement inspection of items before preparation for delivery. (See 4.3)

(2) Procurement inspection of preparation for delivery. (See 4.23 and section 5.)

4.2 First Article Inspection. - This inspection will be performed by the contractor unless otherwise specified in the contract. It shall consist of the First Article inspection specified in Table III and the Group A, Group B, and Group C inspection specified in Tables IV, V, and VI, respectively. Other nondestructive tests on First Article samples may be performed to determine compliance with specified requirements. The First Article inspection will normally be performed in this order: (1) bounce, (2) shock, drop, and (3) immersion; other First Article inspection may precede, follow, or be interspread between the foregoing.

TABLE III Preproduction inspection.

Inspection	Requirement Paragraph	Inspection Paragraph
Bounce	3.15	4.8
Moisture	3.14.1	4.13
Radio interference suppression	3.10.1	4.16
Shock; drop	3.15	4.18
Temperature	3.14.2	4.19

4.3 Procurement inspection before preparation for delivery.- The contractor shall perform the inspection specified by (a) through (c) below, to demonstrate compliance with specified requirements. However, this does not relieve the contractor of responsibility for performing

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any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements. The Government will review and evaluate the contractor's inspection procedures and examine the contractor's inspection records. In addition the Government - at its discretion - may perform all or any part of the inspection specified in (a), to verify the contractor's compliance with specified requirements. (The amount of such verification inspection will depend on the extent to which the contractor's inspection precedures and records insure that the equipment on contract meets the specified procurement inspection. Also see 6.6).

(a) Procurement inspection shall consist of Group A, Group B and Group C inspection as specified in 4.3.1 through 4.3.5.

(b) When Standard MIL-STD-105 specifies actions by the Government, the Government may authorize the contractor to perform any of such actions except that responsibility for acceptance rests with the Government.

(c) Group B inspection shall normally be performed on inspection lots that have passed Group A inspection and on samples selected from units that have been subjected to and met the Group A inspection. In addition, Group C inspection shall normally be performed on sample units that have been subjected to and met Group B inspection. However, the order may be varied when it is considered more practical to select separate samples for Group B or Group C inspection, or both.

4.3.1 Group A inspection.- This inspection (including sampling) shall conform to Table IV and Standard MIL-STD-105. Unless otherwise specified, normal inspection shall be used at the start of the contract. Group A inspection shall be performed in any order which is satisfactory to the Government, except that the air-seal test (4.5) shall be next to last and the operational inspection (4.15) shall be last. (See 6.4)

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TABLE IV Group A inspection

Inspection	Rqt Para	Insp Para	AQL Major	AQL Minor
Visual and mechanical	3.18	4.21		
Tripod MT-1643( )/U			1%	4%
Crank GC-7( )			1%	4%
Generator Unit			1.5 DPHU	6.5 DPHU
Air seal test	3.13	4.5	1%	
Electrical			*	
Breakdown voltage	3.8.7	4.9		
Efficiency **	3.8.5	4.10		
Output rating	3.8.1	4.14		
Radio operation	3.10	4.15	2.5 DPHU for the group	
Ripple voltage	3.8.4	4.17		
Voltage regulation	3.8.2	4.22		

* All electrical defects are considered major.
** Efficiency test under above noted inspection (for production testing) required to be performed at normal ambient temperature only.

4.3.2 Group B inspection.- This inspection (including sampling) shall conform to Table V and the Appendix to Standard MIL-STD-105. Unless otherwise specified herein, normal inspection shall be used at the start of the contract. The reduced inspection procedure shall be R-1. Disposition of nonconforming product (sample units and inspection lots) shall be in accordance with 4.3.5 and the requirements of Standard MIL-STD-105 for disposition of rejected product.

4.3.2.1 Group B sampling plans.- The Group B sampling plans, as listed in Table V shall be as follows:

AQL	Inspection level for normal inspection	Inspection level for reduced inspection
4.0%	L-8	L-6
6.5%	L-7	L-5
10.0%	L-6	L-4
15.0%	L-5	L-3

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4.3.2.2 Order of inspection within Group B.- Group B inspection shall be performed in any order which is satisfactory to the Government.

TABLE V Group B inspection

Inspection	Req Para	Insp Para	AQL
Immersion	3.14.1	4.11	6.5%

4.3.3 Group C inspection.- This inspection shall be as listed in Table VI.

4.3.3.1 Sampling for inspection of equipment.- Sampling for each Group C inspection shall be as specified in Table VI.

4.3.3.2 Noncompliance.- If a sample unit fails Group C inspection, the contractor shall immediately investigate the cause of failure and shall report to the Government inspector the results thereof and details of the corrective action taken on the process and all units of product which were manufactured with the same conditions, materials, processes, etc. If the Government inspector does not consider that the corrective action will enable the product to meet specified requirements, or if the contractor cannot determine the cause of failure, the matter shall be referred to the contracting officer. (See 6.5)

TABLE VI Group C inspection

Inspection	Rqt Para	Insp Para	Sampling plan
Construction Verification	3.1	4.4	
Acoustic noise	3.9	4.7	1 unit from each 100 units produced or fraction thereof
Life	3.11	4.12	1 unit from each months production
Temperature rise	3.8.6	4.20	1 unit from each 250 units produced or fraction thereof

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4.3.4 Reinspection of conforming Group B and Group C sample units.- Unless otherwise specified, sample units which have been subjected to and passed Group B or Group C inspection, or both, may be accepted on contract, provided that they are resubjected to and pass Group A inspection after repair of all visible damage and replacement of all worn parts.

NOTE: Units which have been subjected to life tests must be completely reworked and made equal to new units.

4.3.5 Dispostion of nonconforming product.- The following shall be suitably tagged or identified by equivalent means to indicate the cause of failure and means employed to correct the fault.

Sample units found defective during Group A, B, or C inspection

Inspection lots which failed during Group A or B inspection.

Product which has been reworked as a result of failure during Group C inspection.

The required information shall be presented to the Government when the product is submitted and shall become the property of the Government.

4.4 Construction verification.- The contractor shall perform an Equipment Verification Review (EVR), consisting of a complete technical audit of the equipment on order against the drawings (see 3.1) cited in the Technical Data Package (as supplemented by all approved engineering changes). The EVR shall consist of the following:

a. An audit to establish that "as-built" the generator, including individual parts, their use in subassemblies, and assemblies (components), and their final installation into the unit, are in accordance with the end product drawings.

b. An audit to establish that "as-built" the generator, including parts, components and their installation, meet the acceptance requirements specified in the Technical Data Package for each part, component, and generator. The audit shall record all acceptance test methods used together with resulting verification test data.

c. Preparation of EVR report(s) containing the following information:

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(1) Identification of parts, components, and details proving compliance with 4.4a and b above.

(2) Discrepancies noted.

(3) Corrective action taken.

d. Two (2) copies of the EVR report shall be submitted no later than 15 days prior to the submission of the first production unit or lot to the government technical organization cited in the contract. The results of the EVR shall be subject to verifiction by authorized government personnel at the time the first unit or production lot is offered for acceptance. Government Verification will be to the depth and extent necessary to demonstrate that the "as-built" hardware is in accordance with the cited drawings. Copies of the EVR reports shall be made available to government personnel during the verification. The contractor shall provide the following:

(1) Segregation of parts, components of generators, to permit reviewing government personnel access for detailed inspection. The contractor shall also provide personnel to disassemble any parts of the generator necessary to permit government review of the generator, its components and their installation, and to reassemble parts after inspection.

(2) Have as required, responsible personnel from each functional department available for discussions in their respective areas.

(3) Adequate administrative support for the EVR.

4.5 Air-seal test.- The equipment shall be opened and closed again in such manner as to break and remake the seal. Immediately thereafter, the equipment as field transported shall be subjected to a vacuum of 1 pound per square inch (1-pound per square inch less than the atmospheric pressure surrounding the equipment) applied to the interior of the transit case or to the interior of the equipment closure when no transit case is provided. The vacuum then shall be valved-off and the interior pressure measured during the ensuing period of 1 minute. During this 1-minute period the decrease in vacuum shall not exceed .01 pound. The gage used for measurement of the vacuum shall be of such accuracy that a difference of .01 pound can be determined readily. (See 3.13)

4.6 Test loads.- The generator shall have a 0.150 ± .020 ampere load applied to the high voltage connections (32 and 24) and a 2.9 ± 0.2 ampere load applied to the low voltage connections (34 and 24) when testing for the following:

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(a)	Overall efficiency	4.10
(b)	Life test	4.12
(c)	Ripple voltage	4.17
(d)	Low voltage change with change of temperature	4.19.1
(e)	Temperature rise	4.20
(f)	Voltage regulation	4.22

4.7 Acoustic noise.- The acoustic noise due to operation of the generator, shall be measured with a sound level meter (type 1151C as manufactured by General Radio Co.) with 70 db and 40 db weighting, in a room having an ambient noise level of not more than 40 db.

Reference level, 0db = 10 to the 16th power watts (acoustic) per square centimeter.

These measurements shall be made at a distance of 3 feet from the top, front edge of the generator to the center of the sound level meter microphone, with a 0.050 ampere load applied to the high voltage connection and a 2.5 ampere load applied to the low voltage connection and the generator operating at a cranking speed of 60 ± 2 rpm, the acoustic noise of the generator shall not exceed the values specified in 3.9.

4.8 Bounce test; equipment under 1000 pounds.- The generating unit, in its case (if any), shall be placed on the table of the Package Tester as made by the L.A.B. Corporation, Skaneateles, New York or equal, and shall be constrained from horizontal motion of more than 2 inches by suitable wooden fences. The package tester shafts in phase, shall be operated at a speed of 285 rpm ± 1 percent, for a total of 3 hours. During each 1/2 hour period of the test, the transit case (with contained equipment) shall rest on a different face.

4.9 Breakdown voltage test.- The generating unit shall be tested to determine compliance with requirements specified in 3.8.7. All a.c. voltages indicated are r.m.s., 60 cycle. All capacitors shall be disconnected while making the breakdown voltage tests.

4.10 Efficiency.- The efficiency of the generator shall be measured as follows: The generator shall be mounted to swing freely on its crankshaft. A beam with an effective leverarm exactly 1 foot long measured to the center of the crankshaft, and at right angles to the side of the generator, shall be attached to that side of the generator where the torque will tend to move the beam upward. In order to measure the correct torque produced by the generator, the generator

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should be so balanced that the arm when horizontal will exert a force due only to the rotation of the generator, and not to any unbalance of the arm or generator. After balancing the generator so that the beam is in a horizontal position, the generator shall be operated with the test loads specified in 4.6 applied to the output and at a speed of 60 ± 2 rpm using a d.c. motor and a suitable gear reduction. The vertical restraining force at the end of the beam shall maintain the beam in a horizontal position; the resulting efficiency shall be as specified in 3.8.5.

$$\text{Percent efficiency of the generator} = \frac{7.04\ W}{F\ N} \times 100$$

where W = Watts output of the generator
F = Pounds restraining force required to prevent rotation of the beam
N = Number of revolutions per minute of the crankshaft

7.04 = Constant

4.11 Immersion test.- The equipment, as prepared for field transportation, shall be immersed to a minimum depth of 3 feet of fresh water for 2 hours. Immediately prior to immersion, the temperature of the equipment shall be 40°F, or more, above the temperature of the water. The tank in which the equipment is immersed shall be of sufficient capacity to maintain the water within ± 2°F of its initial temperature, or the temperature of the water shall be maintained within those limits by other means. After completion of the 2-hour period of immersion, the equipment shall be removed from the water and wiped dry on exterior surfaces. When the equipment is opened, there shall be no evidence of leakage.

4.12 Life.- The equipment shall be subjected to the following life test for a total of 1000 hours. The generator shall be connected to resistances, the values of which will simulate full load, and shall be driven at 60 ± 2 rpm by a motor drive. The driving power shall be on continuously and the output loads shall be automatically switched on and off, to give a cycle consisting of 2 minutes on and 4 minutes off. Maintenance cleaning on the generator shall be permitted at a 100 or more hour intervals. Performance following this test shall be as specified in 3.11.

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4.13 Moisture-resistance test for equipment.- The equipment shall be tested as follows:

4.13.1 Test conditions.-

(a) Do not move equipment from the humidity chamber for measurements.

(b) Complete measurements as rapidly as possible.

(c) Perform last two cycles with the cover removed from the generator.

4.13.2 Test method.-

(a) Dry at 130° ± 5°F for 24 hours.

(b) Condition at 77° ± 5°F and 40 to 50 percent relative humidity for 24 hours.

(c) Take measurements as specified in 4.13.3, and readjust as necessary to meet full specification and requirements.

(d) Subject to continuous cycling for five 48-hour cycles. Temperature, relative humidity, and period of time for each portion of the cycle shall conform to Standard MIL-STD-170.

(e) After cycling has been completed and immediately after removal of the equipment from the humidity chamber, remove surplus water with a rag or air hose and within 30 minutes, measure insulation resistance of windings. The minimum resistance of any winding shall be one (1) megohm.

(f) Following performance of insulation resistance measurements specified in (e) above condition the equipment for 24 hours at 77°F ± 5°F and 40 to 60 percent relative humidity. No repair or replacement of parts shall be made. After conditioning, the equipment shall meet full performance for those measurements specified in 4.13.3.

4.13.3 Performance.- Measurements made prior to and after the humidity test (4.13.2) shall consist of the following:

Paragraph	Measurement	Requirement Paragraph
4.10	Efficiency	3.8.5
4.14	Output rating	3.8.1
4.15	Radio operation	3.10
4.17	Ripple voltage	3.8.4

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4.13.4 Failure.- If any equipment fails to meet the minimum insulation resistance specified in 4.13.2(e) above, or fails to meet any of the tests of 4.13.3 as referenced in 4.13.2(f), it does not pass the test.

4.14 Output test.- The equipment shall be tested to determine compliance with the requirements specified in 3.8.1. The generator shall be operated continuously at the specified speed range for not less than 15 minutes at each of the four specified outputs. During this time the output shall remain within specified limits.

4.15 Radio operation test.- The generator shall be connected to radio Receiver-Transmitter RT-77( )/GRC-9 and driven mechanically or manually at rated cranking speed. A power output meter shall be connected to one of the phone jacks of the receiver. The following tests shall be performed in a shielded or screen room.

4.15.1 Radio receiver noise test.- The tuning dials of Receiver-Transmitter RT-77( )/GRC-9 shall be adjusted for 4.8 megacycles with the volume and RF gain controls adjusted to maximum. The antenna selector switch shall be in position 1. The receiver and transmitter selector switches shall be adjusted to PHONE, HIGH SENSITIVITY AND SEND positions. The power output meter reading shall not exceed the value specified in 3.10 for no antenna connection. An antenna, the length of the screen room, shall then be connected to the antenna binding post of the receiver. The power output meter reading shall not exceed the value specified in 3.10 for the antenna connection. The output power meter used shall be Type 538-A as made by General Radio Company, or equal. In making the test, check the set screw in the base of the receiver to determine that the impedance at which it is operated matches the impedance of the output power meter. The impedance of the output meter shall be set at 4000 ohms.

4.15.2 Radio receiver sensitivity test.- The shielded output lead of a signal generator shall be connected to the antenna binding post of Receiver-Transmitter RT-77( )/GRC-9 through a 110-micromicrofarad capacitor, and the shielding shall be connected to the radio housing. The signal generator shall be adjusted for an output signal of 6 microvolts, 30 percent modulated, 400 cycles, at the same radio frequency as the radio receiver frequency. With the signal generator adjusted to zero modulation and the RF gain control at maximum, the radio receiver volume control shall be adjusted to give a power output meter reading of not more than 1 milliwatt. With the signal generator adjusted for a 30 percent modulated signal, the power output meter shall read not less than 10 milliwatts.

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4.15.3 Radio transmitter noise test.- A dummy load, consisting of a 0-1 ampere R.F. ammeter in series with a 20-ohm noninductive resistor, and a 20-micromicrofarad capacitor, shall be connected between the antenna binding post and the metal housing of Receiver-Transmitter RT-77( )/GRC-9. The transmitter frequency shall be adjusted to 4.8 megacycles. The antenna selector switch, and the antenna tuning control shall be adjusted for highest selector switch, and the antenna tuning control, shall be adjusted for highest antenna current. A second Receiver-Transmitter RT-77( )/GRC-9 shall be tuned to the transmitter signal, and the signal listened to; there shall be no evidence of warble, ripple, hash, or distortion. This test shall be conducted with the transmitter adjusted for HIGH power operation on CW, MCW, and PHONE positions.

4.16 Electromagnetic Interference (EMI).- The equipment shall comply with the requirements of Para 3.10.1, using the test methods and procedures of Notice 3, MIL-STD-462. The equipment shall be tested to Method RE 02 in a portable configuration and shall be driven remotely during the test at a rotational speed of 60 ± 2 rpm. The driving source and the load for the equipment shall be external to the shielded inclosure. (See 3.10.1)

4.17 Ripple voltage.- Both the high voltage and the low voltage outputs shall be tested for ripple content using a device such as a Ballatine Model 300A vacuum tube voltmeter, or equal, shunted with a 20,000 ohm resistor, and connected to the generator through a 4 microfarad nonelectrolytic capacitor. The generator shall be operated at a cranking speed of 60 ± 2 rpm under the test loads specified in 4.6 and in the temperature range specified in 3.8.4. The percent ripple shall not exceed the value specified in 3.8.4 and shall be determined by the following equation:

$$\text{Percent ripple} = \frac{E1}{E2} \times 100$$

where E1 = ripple voltage component and E2 = dc output voltage contained in the ripple E1

4.18 Shock test; drop.- The equipment shall be dropped a distance of 3 feet on each corner, edge, and face (total of 26 drops). Drops shall be made from a quick-release hook; an Acme Drop Tester, or equal; a Conbur Tester (10-degree incline plane), or equal; or a basket-type pendulum tester. The floor or barrier receiving the impact shall be 2 inch fir, backed by concrete or a rigid steel frame.

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4.19 Temperature test for equipment.- The equipment shall be subjected to the temperature cycle shown on Standard MIL-STD-169. Measurements shall be taken as follows:

Paragraph	Step	Measurement	Test results
4.10	4 and 8	Efficiency	Full specification requirements
4.14	4 and 8	Output rating	Full specification requirements
4.15	4 and 8	Radio operation	Full specification requirements

4.19.1 Low voltage change with change of temperature.- The temperature shall be cycled in accordance with Standard MIL-STD-169. The equipment shall be tested at steps 4 and 8 to determine compliance with requirement specified in 3.8.3.

4.20 Temperature rise test.- The temperature rise due to the operation of the assembled generator shall be measured as follows: The thermocouple of a pyrometer shall be attached to the surface of the generating unit. The temperature of the generator shall be approximately equal to the ambient temperature as specified in 3.8.6. The generator shall then be operated continuously under test loads specified in 4.6 at a cranking speed of 60 + 2 rpm in still air. After 30 minutes of operation, the temperature shall be measured. The rise in temperature of the generator above the ambient temperature shall not exceed the value specified in 3.8.6. Ambient temperature shall be determined by taking temperature measurements in still air at six points in the same plane radially about the equipment at a distance not exceeding 2 feet from the equipment.

4.21 Visual and mechanical inspection.- Equipment shall be examined for the defects listed in Standard MIL-STD 252 and Table VII.

TABLE VII - Classification of visual and mechanical defects.

Classification	Defect
Major	Improper fit of Crank GC-7( ) Improper mounting of generator on tripod. Insufficient clearance of mating gears Improper alignment of shafts, gears,and bearings. Cracks in cover, etc.
Minor	Dents Improperly cleaned units Illegible lettering and numbering Defective finish

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4.22 Voltage regulation.-

4.22.1 High voltage.- The high voltage output shall be measured when the generator is operating at a cranking speed of 60 ±2 rpm under the test loads specified in 4.6. The load shall then be removed and the high voltage output measured again. The percent of high voltage regulation shall not exceed the value specified in 3.8.2.1 and shall be determined by the following equation:

$$\text{Percent regulation} = \frac{E1-E2}{E1} \times 100$$

where E1 = voltage at no load and E2 = voltage at full load.

4.22.2 Low voltage.- The low voltage output shall be measured when the generator is operating at a cranking speed of 60 ± 2 rpm with the high voltage output unloaded and a 2.9 ampere load applied to the low voltage connection. The low voltage load shall then be removed and output measured again. The percent of voltage regulation shall not exeeed the value specified in 3.8.2.2 and shall be determined by the equation contained in 4.22.1.

4.22.3 Change of Speed.- The output voltages shall be measured when the generator is operating at a cranking speed of 60 ± 2 rpm under test loads specified in 4.6. The cranking speed shall then be changed to 50 ± 2 or 70 ± 2 rpm and the output voltages measured again. The percent of voltage regulation shall not exceed the value specified in 3.8.2.3 and shall be determined by the following equation:

$$\text{Percent regulation} = \frac{E1-E2}{E1} \times 100$$

where E1 = voltage at 60 rpm and E2 = voltage at 50 or 70 rpm

4.23 Procurement inspection of preparation for delivery.- The contractor shall perform the inspection specified in 4.23.1 and 4.23.2, to demonstrate compliance with specified requirements. However, this does not relieve the contractor of responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements. The Government will review and evaluate the contractor's inspection procedures and examine the contractor's inspection records. In addition the Government--at its discretion--may perform all or any part of the

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inspection specified in 4.23.1 and 4.23.2, to verify the contractor's compliance with specified requirements. (The amount of such verification inspection will depend on the extent to which the contractor's inspection procedures and records insure that the equipment on contract meets the specified procurement inspection. Also see 6.6). When Standard MIL-STD-105 specifies action by the Government, the Government may authorize the contractor to perform any of such actions except that responsibility for acceptance rests with the Government.

4.23.1 Preservation and packaging.- Inspection of preservation and packaging shall be as specified in Specification MIL-P-116. Classification of defects shall be as shown on Table VIII.

TABLE VIII

PRESERVATION AND PACKAGING AND MARKING THEREOF

MAJOR	MINOR
1. Use of improper or defective material.	1. Item not properly blocked or braced within the unit package to prevent movement.
2. Quantity in unit package not as specified.	2. Packaging material damaged.
3. Incorrect packaging method applied.	3. Any item of marking information other other than 7 thru 9 omitted, incorrect, or illegible.
4. Cushioning or padding omitted.	
5. Cushioning inadequate for the physical and mechanical protection of the item.	
6. Unsealed, punctured, or improperly sealed barrier bag, wrap, or envelope.	
7. Stock number omitted, incorrect, or illegible.	
8. Nomenclature omitted, incorrect, or illegible.	
9. Marking of quantity of items in package omitted, incorrect or illegible.	
10. Different stock numbered items in the same unit package.	

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4.23.2 Packing and marking of exterior containers.- Packing and marking of exterior containers shall be given visual inspection for the defects listed below and to determine conformance with the approved process sheet furnished by the contractor as required by the contract. This inspection shall conform to the Appendix to Standard MIL-STD-105. Inspection level L-8 shall be used for normal inspection and L-6 for reduced inspection. Unless otherwise specified herein, normal inspection shall be used at the start of the contract. The reduced inspection procedure shall be R-1. The AQL for major defects shall be four percent and the AQL for minor defects shall be ten percent. Classification of defects shall be as shown on Table IX.

TABLE IX

PACKING AND MARKING THEREOF

MAJOR	MINOR
1. Use of improper or defective material.	1. Unsealed carton.
2. Quantity in pack not as specified.	2. Defective taping or sealing of carton.
3. Gross weight in excess of specified amount.	3. Any other box defect which may be considered minor by definition of Standard MIL-STD-105.
4. Box closure not as specified.	3. Any item of required marking information other than 10 thru 16 listed under major defects omitted, incorrect, or illegible.
5. Type, grade, class and style of the shipping container not as specified.	
6. Strapping omitted.	
7. Strapping inadequate or incorrectly applied.	
8. Items not adequately blocked, braced or cushioned within the shipping container to prevent movement or damage.	
9. Shipping documents or packing list omitted.	
10. Stock number omitted, incorrect, or illegible.	
11. Nomenclature omitted, incorrect, or illegible.	

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12. Marking of quantity of items in pack omitted, incorrect, or illegible.

13. Destination marking omitted, incorrect, or illegible.

14. Service designation (color marking) omitted.

15. Specified special marking and labeling such as NAP labels, shipment digit markings, etc., omitted, incorrect, or illegible.

16. Overseas code marking omitted, incorrect, or illegible.

4.23.2.1 Inspection lot.- A lot for visual inspection of the pack shall be all completed packs which are identical and will be submitted for acceptance at one time.

4.23.2.2 Disposition of nonconforming product.- Disposition of nonconforming product (sample units and inspection lots) shall be in accordance with the requirements of Standard MIL-STD-105 for disposition of rejected product. When submitted for acceptance, such product shall be suitably tagged or identified by equivalent means to indicate the cause of failure and means employed to correct the fault. The required information shall be presented to the Government when the product is submitted and shall be come the property of the Government.

5. PACKAGING

5.1 Preservation.- Preservation shall be level A, B or commercial, as specified (see 6.2(b)).

5.1.1 Level A

5.1.1.1 Cleaning.- Generator, Direct Current G-43( )/G (hand cranked) shall be cleaned in accordance with process C-1 of MIL-P-116.

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5.1.1.2 Drying.- Generator, Direct Current G-43( )/G (hand cranked) shall be dried in accordance with the applicable procedure of MIL-P-116.

5.1.1.3 Preservative application.- None required.

5.1.1.4 Unit packing.- Unit packing shall be in accordance with the methods prescribed in MIL-P-116 as specified herein.

5.1.1.4.1 Generator, Direct Current G-43( )/G (hand cranked).- Each generator shall be individually unit packed in accordance with items 1 through 14 of Table X and Figures 1 and 2.

5.1.2 Level B.- Cleaning, drying, preservative application and unit packing shall be as specified in 5.1.1.

5.1.3 Commercial preservation.- Preservation shall be in accordance with MIL-STD-1188.

5.2 Packing.- Packing shall be level A, B or commercial, as specified (see 6.(b)) Shipping containers for level A and B shall be capable of stacking and supporting superimposed loads during shipment and storage without damaging the container(s) or its contents.

5.2.1 Level A.- A quantity of Generator, Direct Current G-43( )/G (hand cranked), unit packed as specified in 5.1, shall be packed within a closefitting box conforming to PPP-B-601, overseas type; PPP-B-621, style 4, class 2; or PPP-B-585, style 2 or 3, class 3. When the gross weight exceeds 200 pounds, or the container length and width is 48 x 24 inches or more and the weight exceeds 100 pounds, 3 x 4 inch skids, laid flat, shall be applied in accordance with the requirements of the container specification, or if not specified in the specification, in a manner which will adequately support the item and facilitate the use of material handling equipment. Closure and strapping shall be in accordance with the applicable container specification or appendix thereto except that metal strapping shall conform to QQ-S-781, type I, finish A.

5.2.1.1 Unitization.- Palletization shall be required when: containers specified in 5.2.1 do not require skids; quantities per destination exceed either a total of 250 pounds (excluding the pallet) or a volume of 20 cubic feet; and the container size permits use of one of the pallet patterns of MIL-STD-147. A quantity of containers, packed as specified in 5.2.1 except that container strapping may be omitted, shall be placed on a pallet, load type I, conforming to MIL-STD-147. The pallet shall conform to NN-P-71, type IV, group I or II woods. The load shall be "bonded" to the pallet by strapping conforming to QQ-S-781, type I, finish A, or shrink film conforming to L-P-378, type IV."

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5.2.2 Level B.- A quantity of Generator, Direct Current G-43( )/G (hand cranked), unit packed as specified in 5.1, shall be packed within a closefitting fiberboard box conforming to PPP-B-640, class 2, style E, or PPP-B-636, type CF, class weather-resistant. The gross weight of boxes conforming to PPP-B-640 shall not exceed 250 pounds. Closure shall be in accordance with the appendix of the applicable box specification. Reinforcing shall be by pressure-sensitive filament tape banding or non-metallic strapping conforming to PPP-T-97 and PPP-S-760, respectively; selection of the material and application shall be in accordance with the appendix of the applicable box specification. When the gross weight exceeds 200 pounds, or the container length and width is 48 x 24 inches or more and the weight exceeds 100 pounds, containers will be pallet-mounted on pallets conforming to NN-P-71, group I or II woods. The load shall be "bonded" to the pallet by strapping conforming to QQ-S-781, type I, finish A or shrink film. When a single unit is shipped to a single destination no packing is required; the unit container shall serve as the shipping container. Reinforcing shall be as specified above.

5.2.2.1 Unitization.- Palletization shall be required when quantities per destination exceed either a total of 250 pounds (excluding the pallet) or a volume of 20 cubic feet; and the container size permits use of one of the pallet patters of MIL-STD-147. A quantity of Generator, Direct Current G-43( )/G (Hand Cranked), unit packed as specified in 5.1, shall be placed on a pallet, load type I, conforming to MIL-STD-147. The pallet shall conform to NN-P-71, type IV, group I or II woods. The load shall be "bonded" to the pallet by strapping conforming to QQ-S-781, type I, finish A, or shrink film conforming to L-P-378, type IV."

5.2.3 Commercial packing.- Packing shall be in accordance with MIL-STD-1188.

5.3 Marking.-

5.3.1 Military marking .- In addition to any special marking required by the contract or order, interior packs and exterior shipping containers shall be marked in accordance with MIL-STD-129.

5.3.2 Commercial marking.- In addition to any special marking required by the contract or order, interior packs and exterior shipping containers shall be marked in accordance with MIL-STD-1188.

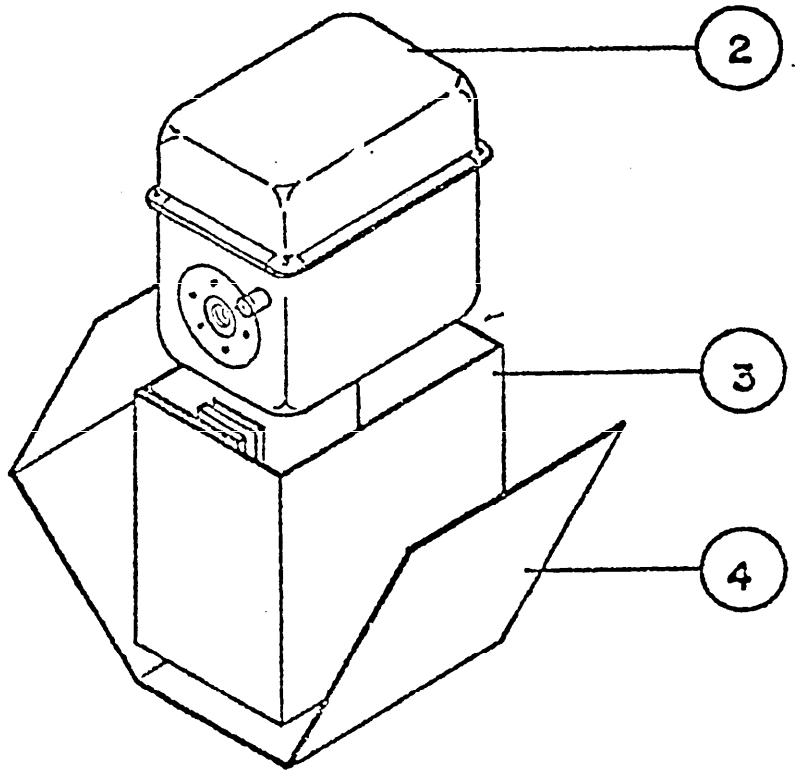
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TABLE X. Bill of material.

REF No.	ITEM Nomenclature	No. Reqd	Method MIL-P- 116	SIZE - INCHES L	W	D	Spec No.	REQUIREMENT T	C	V	G	S
1	Technical Literature	2	IC-1	11	8							
	Bag	2		14	10		MIL-B-117	I				1
2	Direct Current Generator G-43( )/G	1	III	9-3/8	6	9						
3	Liner, Fiberboard	1		9-3/8	6	9-3/8	PPP-F-320	CF	WR	SW	W5c	
4	Liner, Fiberboard	1		9-1/2	6-1/8	9-1/2	PPP-F-320	CF	WR	SW	W5c	
5	Cushioning Material	1		10	7	1/2	PPP-C-843	II	B			
6	Cranks	2	III									
	Tape	2		16	2		PPP-T-45	III				
7	Box, Fiberboard	1		9-3/4	6-1/2	10	PPP-B-636				W5c	
	Box closure shall be as specified in the appendix of the box specification.											
8	Seat and Tripod Assembly	1	III									
	Extend seat to extreme position, leaving legs together.											
9	Strip, Fiberboard	1		72	6-3/4		PPP-F-320	CF	WR	SW	W5c	
10	Canvas, Cover	1	IC-1									
	Bag	1		24	12		MIL-B-117	I	C			1
11	Cell, Fiberboard (triangle)	1		6	4-1/2	1-1/2	PPP-F-320	CF	WR	SW	W5c	
12	Cushioning Material	1		24	7	1/2	PPP-C-843	II	B			
13	Cell, Fiberboard	1		50	6-3/4		PPP-F-320	CF	WR	SW	W5c	
14	Box, Fiberboard	1		34	6-3/4	12-7/8	PPP-B-636				W5c	
	Box closure shall be as specified in the appendix of the box specification.											

T - Type C - Class V - Variety G - Grade S-Style

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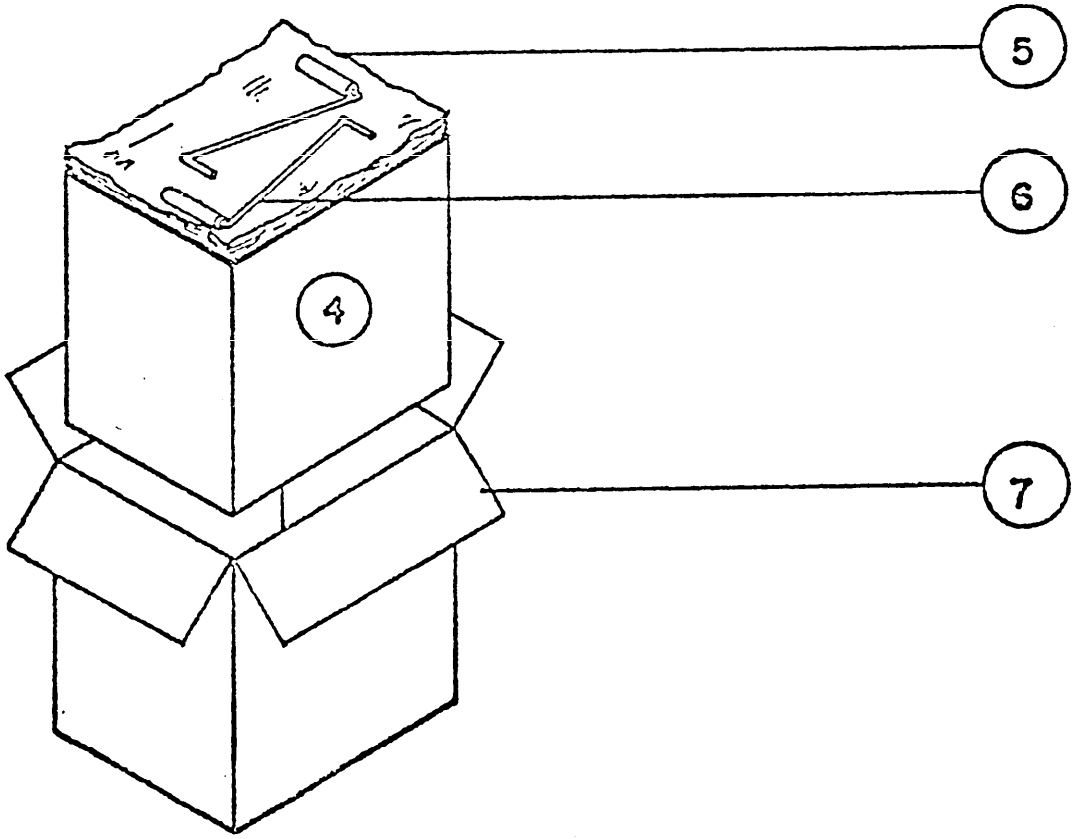


FIGURE 1. Generator, Direct Current G-43( )/G (hand cranked).

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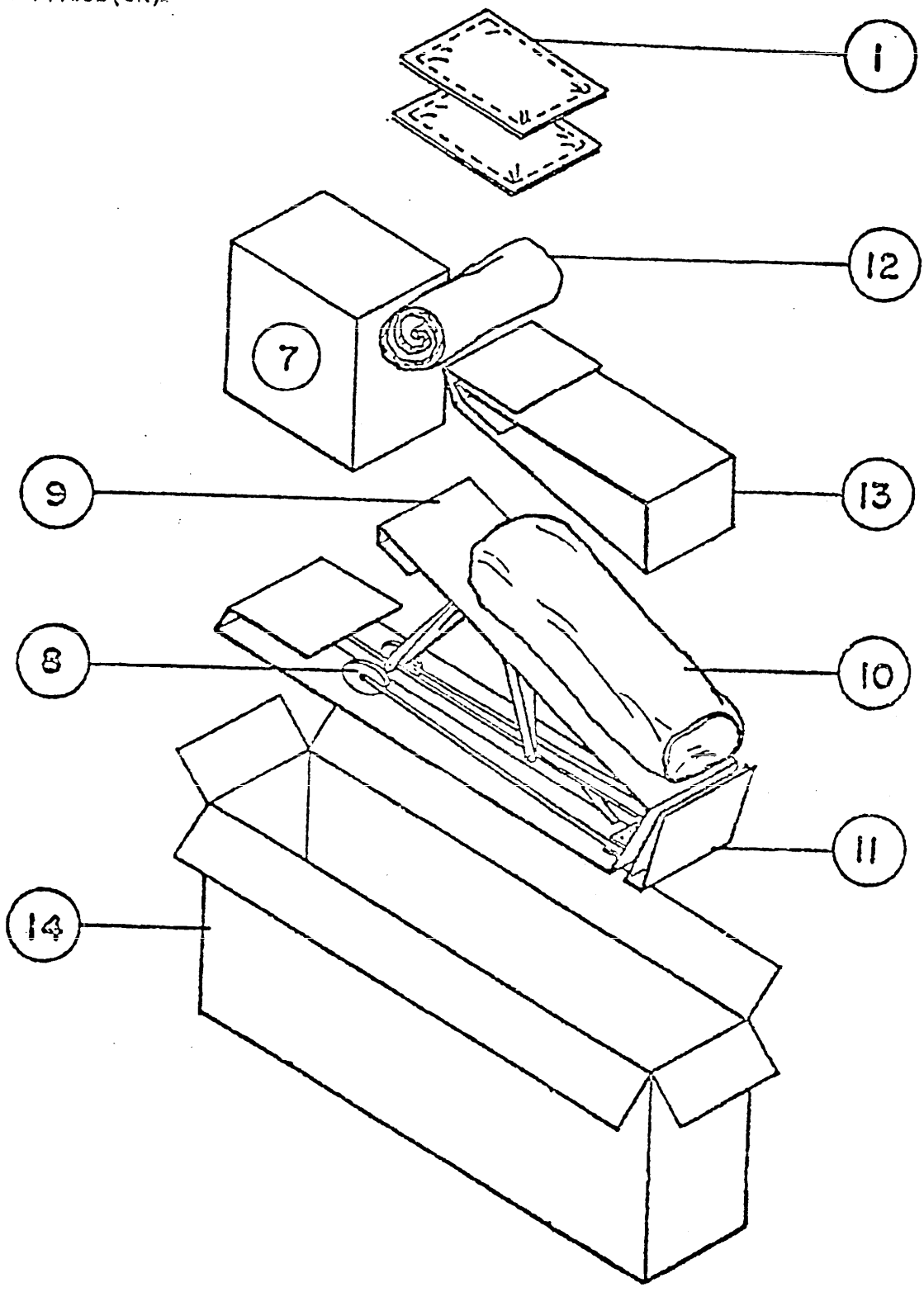


FIGURE 2. Generator, Direct Current G-43( )/G (hand cranked). - Continued

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6. NOTES

6.1 Intended use.- Generator G-43( )/G shall be used to furnish power to Receiver-Transmitter RT-77( )/GRC-9.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number, and date of this specification and any amendment thereto.

(b) Level A, level B or commercial preservation and packing (see section 5).

(c) First Article Inspection: Two samples of each item cited in section 1 are generally required so that lengthy environmental tests can be completed on one sample while complete performance measurements can be made on the second sample (See 3.16)

6.3 Nomenclature.- The parentheses in the nomenclature will be deleted or replaced by a letter identifying the particular design; for example: G-43W/G. The contractor should apply for nomenclature in accordance with the applicable clause in the contract.

6.4 Location of operational inspection and air-cool test.- It is desirable that the operational inspection (4.15) and the air-seal test (4.5) be performed at a location that will minimize handling (which might cause damage to the equipment) after this inspection is completed. Any preparation for shipment which would require breaking of the equipment seal should be accomplished prior to the air-seal test so that the seal may remain intact thereafter. It is recommended that the entire lot (including all previously inspected sample units) be sampled and inspected immediately prior to packaging.

6.5 Group C inspection.- Approval to ship may be withheld, at the discretion of the Government, pending the decision from the contracting officer on the adequacy of corrective action. (See 4.3.3.2)

6.6 Verification inspection.- Verification by the Government will be limited to the amount deemed necessary to determine compliance with the contract and will be limited in severity to the definitive Quality Assurance Provisions established in this specification and the contract The amount of verification inspection by the Government will be adjusted to make maximum utilization of the contractor's quality control system and the quality history of the product, and will normally be identified by the categories listed below:

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(a) Type A--The total of that inspection set forth in the Quality Assurance Provisions of this specification of the contract. Included in this category is that amount of inspection referred to as normal and tightened inspection by Military Standard 105.

(b) Type B--That inspection set forth in the Quality Assurance Provisions of this specification or the contract reduced in amount under the reduced inspection provisions of Military Standard 105.

(c) Type C--A reduced inspection procedure resulting in a material reduction in the amount of inspection set forth in the Quality Assurance Provisions of this specification. The amount of inspection is less than that provided for in type B and is based upon a consistently acceptable product resulting from a planned quality control system voluntarily employed by the contractor in the production of the product.

6.7 Dimensional data.- Dimensions of packaging materials in the Bill of Materials Table are based on the dimensions of the equipment cited in the table. When the dimensions of the equipment vary from those cited, the dimensions of packaging materials shall be adjusted accordingly.

6.8 Environmental.- Environmental pollution prevention measures are contained in the packaging material specifications referenced herein. Refere to material specificatons or preparing activity for recommended disposability methods.

Custodian:
Army - CR

Preparing Activity:
Army - CR

Project No. 6115-A446

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

MIL-A-52878B Alarm-Monitor Group

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426

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